13.2 Derivatives and Integrals of Vector Functions

Prerequisite information for Section :

Give the limit definition of the derivative of a function of one variable (as in section 2.8 of your text), and explain when it is valid (when it applies and when it doesn't).

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

It's valid for all x where the limit exists and $\lim_{x \to a} f(x) = f(a)$

Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Tangent vector:** [after box #1 and before Figure 1 of Section 13.2]

  The vector tangent to the curve at a point.

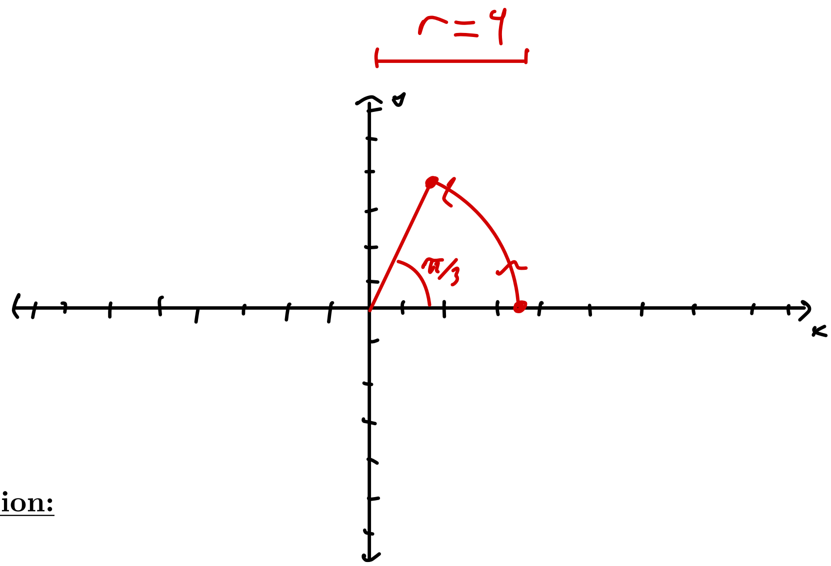
- **Definite integral of a vector function (in three dimensions):** [in unnumbered box before Example 4 of Section 13.2]

$$\int_a^b r(t)\,dt = \left(\int_a^b f(t)\,dt\right)\hat{i} + \left(\int_a^b g(t)\,dt\right)\hat{j} + \left(\int_a^b h(t)\,dt\right)\hat{k}$$

13.3 Arc Length and Curvature

Prerequisite information for Section 13.3: Using methods from Calculus 2 (MAT 192), calculate the arclength of the parametrically defined curve, $x = 4\cos t$, and $y = 4\sin t$, where $t \in [0, \pi/3]$. Show all your work, and give an exact answer. (You should not need your calculator.) After calculating the arclength, describe and/or accurately sketch the given parametrized curve. (Hint: You may wish to review sections 10.1 and 10.2 of your text.)

$$\int_0^{\pi/3} \sqrt{16\sin^2(t) + 16\cos^2(t)}\, dt = 4t\Big|_0^{\pi/3} = 4\pi/3$$



Definitions in this section:

- **Arclength function s (in three dimensions):** [Box # 6 of section 13.3]

For $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$, $t \in [a, b]$

$$\frac{ds}{dt} = |\vec{r}\,'(t)|$$

$$s(t) = \int_a^t |\vec{r}\,'(u)|\, du = \int_a^t \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2 + \left(\frac{dz}{du}\right)^2}\, du$$

- **A smooth parametrization (of a curve):** [discussion before Figure 4 of Section 13.3]

$\vec{r}(t)$ is smooth for an interval $t \in I$ if $\vec{r}\,'$ is continuous and $\vec{r}(t) \neq \vec{0}$ on I.

- **Curvature:** [give the definition from Box # 8 of Section 13.3; this is the *intuitive but useless* definition]

$$\kappa = \left|\frac{d\vec{T}}{ds}\right|$$

- **Principal unit normal vector (or unit normal):** [in discussion before Figure 6 of Section 13.3]

If $|\vec{r}(t)| = c \in \mathbb{R}\ \forall t$, then $\vec{r}(t)\cdot\vec{r}(t) = |\vec{r}(t)|^2 = c^2 \Rightarrow \frac{d\vec{r}}{dt} = 0 = 2\vec{r}'(t)\cdot\vec{r}(t)$

$\Rightarrow \vec{r}'(t)\cdot\vec{r}(t) = 0$

Notice $|\vec{T}(t)| = 1\ \forall t$, thus $\vec{T}'(t)\cdot\vec{T}(t) = 0$. The unit normal is

given as $\vec{N}(t) = \frac{\vec{T}'(t)}{|\vec{T}'(t)|}$ since $|\vec{T}'(t)| \neq 1$

- **Binormal vector:** [in discussion before Figure 6 of Section 13.3]

$$\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$$

- **Torsion (of a curve):** [mathematical definitions given in Box # 13 and Box # 14, but I recommend the intuitive definition given below these boxes]

$$\tau - \frac{d\vec{B}}{ds}\cdot\vec{N}$$

$$\tau(t) = -\frac{\vec{B}'(t)\cdot\vec{N}(t)}{|\vec{r}'(t)|}$$

Topics to learn in this section:

1. The arc length and curvature formulas.
2. The independence of arc length and parametrization.
3. The geometric definition of curvature.
4. The TNB frame.

13.4 Motion in Space: Velocity and Acceleration

Prerequisite information for Section 13.4: Calculus was "invented" to solve two problems, one of which was the analysis of motion. Given that a particle is moving along a straight line with velocity $v(x) = 3x^2 + e^{2x}$, give a *plausible* position function for the particle. (There are infinitely many correct answers; **choose one, specific answer**.)

$$r(x) = \int v(x)\,dx = \boxed{x^3 + \frac{e^{2x}}{2}}$$

Definitions in this section: Position function: $\vec{r}(t)$

- **Velocity vector:** [Box # 2 of Section 13.4 and discussion surrounding this box]

$$\frac{\vec{r}(t+h) - \vec{r}(t)}{h}$$

- **Speed:** [after Figure 1 of Section 13.4]

$$|\vec{v}(t)|$$

- **Acceleration:** [after Figure 1 of Section 13.4]

$$\vec{a} = \vec{v}'(t) = \vec{r}''(t)$$

- **Newton's Second Law of Motion:** [before Example 4 of Section 13.4]

If a force $\vec{F}(t)$ acts on an object of mass m producing acceleration $\vec{a}(t)$, then $\vec{F}(t) = m\vec{a}(t)$

Topics to learn in this section:

1. Definitions of velocity and acceleration as vector functions.
2. How to derive (anti-derive) velocity from acceleration and position from velocity.
3. Tangential and normal components of acceleration.
4. **Note:** Discussion on Kepler's Laws is optional . . . *and we will be following the option to not cover this material in class.*